

Term paper 2

BAN403

Candidate 74 &

April 14, 2026

NHH



Contents

1	Input analysis	2
1.1	Purpose	2
1.2	Inputs	2
1.3	Extracting relevant data	3
1.4	Fitting appropriate statistical distributions	4
1.5	Goodness-of-fit test	5
1.6	Discussing input assumptions	5
1.7	Final input model	5
2	Output Analysis	7
2.1	Design	7
2.2	Performance measures	7
2.3	Confidence intervals	7
2.4	Statistical comparison	7

1 Input analysis

1.1 Purpose

This section provides details for the stochastic inputs for the simulation in accordance with chapter 9 of the textbook (Laguna & Marklund, 2019). The purpose is to identify different elements of the Miller Pain Treatment Center (Chambers & Williams, 2017), organize the available data, fit appropriate probability distributions and justify all assumptions made in the process.

1.2 Inputs

The inputs from the Miller Pain Treatment Center can be categorized into three main groups; patient arrivals, treatment times and outcomes for patients (routing decisions). Each of these groups contains multiple stochastic variables that must be modelled probabilistically to capture the random nature of the system.

Patient arrivals are stochastic as they depend on patients arriving early, on time or late for their appointments. This results in queues and system congestion, which must be captured by the model.

Treatment times also vary between patients depending on their conditions and the complexity of the treatment needed. These variables include registration time, vitals, consultation time and treatment, as well as time spent on teaching residents.

Finally, patient outcomes provides stochastic routing decisions in the system. According to their needs, patients follow different paths through the system, resulting in varying resource utilization, waiting times and overall system performance.

All of these variables must be captured and modelled probabilistically to ensure the simulation accurately depicts the system. Table 1 summarizes the stochastic variables identified in the Miller Pain Treatment Center, categorized into the three main groups:

Variable	Type	Description
Patient arrivals		
Arrival deviation	Continuous	Difference between scheduled and actual arrival times
Treatment times		
Registration time	Continuous	Time spent on administrative registration
Vitals time	Continuous	Time to record patient vitals
Physician assistant time	Continuous	Time spent by PA on follow-up patients
Check-out time	Continuous	Time to complete visit and paperwork
Resident review (new)	Continuous	Time reviewing patient file (new patients)
Resident–patient interaction (new)	Continuous	Consultation time with resident
Teaching time (new)	Continuous	Teaching interaction between resident and attending
Resident–attending interaction (new)	Continuous	Discussion between resident and attending
Resident review (return)	Continuous	Time reviewing file (return patients)
Resident–patient interaction (return)	Continuous	Consultation time with resident
Teaching time (return)	Continuous	Teaching interaction for return patients
Resident–attending interaction (return)	Continuous	Discussion between resident and attending
Patient outcomes (routing decisions)		
Patient type	Discrete	New, return, or follow-up patients
Follow-up requires attending	Discrete	Whether PA consults attending (approximately 50%)
No-show / cancellation	Discrete	Whether patient attends appointment (approximately 10% no-show)
Resident availability	Discrete	Number of residents available per session

Table 1: Identified stochastic inputs

1.3 Extracting relevant data

The data used in the model is based on the data provided by the Miller Pain Treatment Center (Chambers & Williams, 2017), with the accompanying data from the spreadsheet given. Only data that affects system performance and queue times is used in the model.

Data for patient arrivals is extracted from the spreadsheet, specifically the deviation between scheduled and actual arrival times. This is essential to capture queue times and congestion in the system, affecting the entire process.

The spreadsheet contains observations for each activity described in the system, with between 90 and 100 observations for each variable. These observations are used to model the service processes in the system.

Finally, relevant routing decisions are also extracted from the case description, to

capture the stochastic nature of patient outcomes and their paths through the system. Patient type (new, return or follow-up), decisions regarding follow-up patients and no-show rates are all extracted to determine how patients flow through the system, affecting resource utilization and waiting times.

1.4 Fitting appropriate statistical distributions

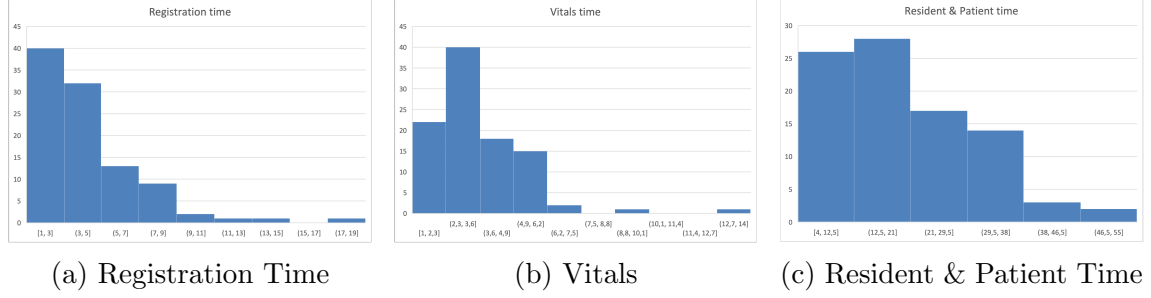


Figure 1: Service times

To determine a suitable probability distribution, histograms can be used to identify the underlying shape of the data, which can be compared to known theoretical distributions (Laguna & Marklund, 2019, p. 362). Figure 1 presents histograms for the three example variables registration time (Figure 1a), vitals (Figure 1b) and resident & patient time (Figure 1c).

All three histograms provide a clear right-skewed form with a long tail towards higher values, while most observations are found at lower values. Although the variation differs for the three variables, the overall shape of the data is consistent. This suggests that a positive and flexible distribution, such as Gamma or exponential could be used. We test for these two, in addition to the Lognormal and Weibull distribution.

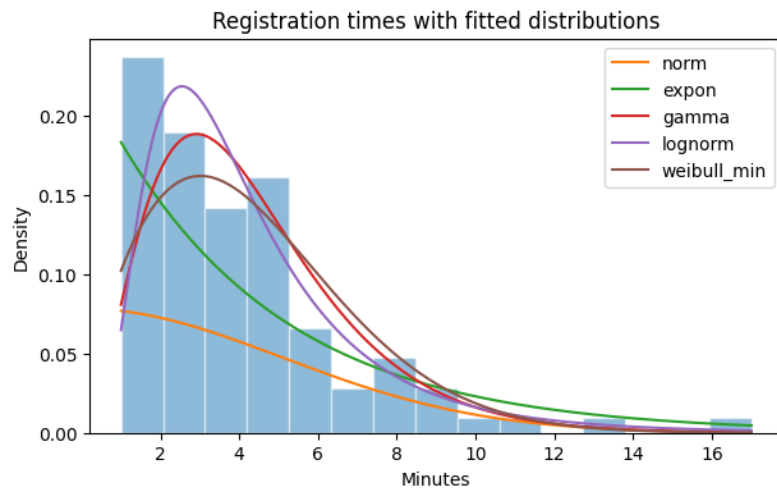


Figure 2: Fitting distributions

Figure 2 shows the histogram of registration times with fitted probability density functions. Using maximum likelihood estimation we can see that Gamma and

Lognorm distributions provide the closest fit to the data, as they capture both the right-skewed nature and long tail of the dataset.

1.5 Goodness-of-fit test

Using a Kolmogorov-Smirnov goodness-of-fit test we can evaluate the suitability of the Gamma distribution. The KS-test measures the largest absolute deviation between the empirical and theoretical cumulative probability distribution functions (Laguna & Marklund, 2019, p. 423). If the deviation is small enough, along with a sufficient p-value, the distribution is considered to be a good suit for the data.

The Gamma distribution is selected as it yields a KS statistic of 0.1132 with a p-value of 0.1466, compared to lognormals KS statistic of 0.1195 and p-value of 0.1089. A significance level above 5% indicates that the null-hypothesis that the data follows a Gamma distribution cannot be rejected, and that the deviation can be explained by randomness (Laguna & Marklund, 2019, p. 425). This is then done for all variables in the system.

1.6 Discussing input assumptions

We make multiple assumptions regarding our dataset to make our model realistic. First, we assume that time is continuous. The dataset provides time in minutes as integers, although it is inherently continuous. This assumption is made to better capture the nature of time, and justifies the use of a continuous distribution such as the Gamma distribution.

As some variables, such as teach and resident review contains zero-values, all distributions are fitted with a location parameter of -0.01. Gamma and Lognormal distribution requires strictly positive values, so this is a technical assumption made to ensure all distributions can be fitted.

Second, we assume that the observations are independent of each other, meaning that one patient and their service times does not affect the next patients service time. It is also assumed that service times are steady throughout the observation period, so that the underlying distribution does not change.

Finally, as all distributions are rejected by the KS test for the variable vitals, we use the best distribution available, which is the weibull distribution. This comes at a KS statistic of 0.2028 with a p-value of 0.0005.

1.7 Final input model

Table 2 provides all variables with their distributions and parameters.

Variable	Distribution	Shape	Scale
Registration	Gamma	3.069	1.415
Vitals	Weibull*	2.107	3.982
Physicians Assistant	Lognormal	0.541	19.463
Check Out	Gamma	2.320	2.041
Resident Review (New)	Gamma	1.401	7.461
Resident & Patient (New)	Lognormal	0.558	17.414
Teach (New)	Gamma	1.879	4.184
Resident & Attending (New)	Gamma	2.831	4.401
Resident Review (Return)	Lognormal	1.002	5.678
Resident & Patient (Return)	Weibull	1.740	14.665
Teach (Return)	Weibull	1.173	5.438
Resident & Attending (Return)	Lognormal	0.738	6.974

* No distribution with $p > 0.05$, Weibull chosen as best fit

Table 2: Final inputs with distributions and parameters

2 Output Analysis

2.1 Design

The Miller Pain Treatment Center is a terminating system, with each session starting empty and ending when all patients have checked out of the system. This means there is no need for a warm-up.

As each session lasts approximately 4 hours (first appointment 08:00, last at 11:15), we set the simulation time to 6 hours to ensure that every patient could finish their visit. All models were run with 30 replications, [INSERT EXPLANATION]. The random seeds used in the probability distributions were consistent for all models, enabling comparison and reducing variance across simulations.

2.2 Performance measures

The primary output variable, as defined by Dr. Weem is mean cycle time per patient (Chambers & Williams, 2017). Mean cycle time is defined as time in minutes from arrival to checkout for all patients per iteration. Patients processed is a secondary concern, meaning the number of patients processed through the system for each replication.

2.3 Confidence intervals

As each replication produces a mean cycle time, 30 iterations provides 30 independent observations of this variable. Since the values vary due to randomness, we make a 95% confidence interval for the sample mean. To calculate this interval we use t-distribution with $n - 1 = 29$ degrees of freedom, as the standard deviation in the population is unknown and therefore estimated from the data. The interval is calculated as $\bar{x} \pm t_{0.025,29} \cdot \frac{s}{\sqrt{30}}$, where $\bar{x}$ is the mean cycle time, s is the standard deviation and $t_{0.025,29}$ is the critical value.

Table here

2.4 Statistical comparison

To compare the output variables from the different scenarios we perform a paired t-test. As all models use the same random seeds, each scenario is exposed to the same set of stochastic inputs, meaning the difference in performance comes from modifications done to the TO-BE models rather than randomness. The difference is defined as $D_i = \bar{x}_i^{TO-BE} - \bar{x}_i^{AS-IS}$, with the hypothesis tested with a significance below 0.05.

References

- Chambers, C., & Williams, K. (2017). Case—Miller Pain Treatment Center. *INFORMS Transactions on Education*, 17(3), 121–127. <https://doi.org/10.1287/ited.2017.0176cs>
- Laguna, M., & Marklund, J. (2019). *Business process modeling, simulation and design* (3rd). CRC Press.